

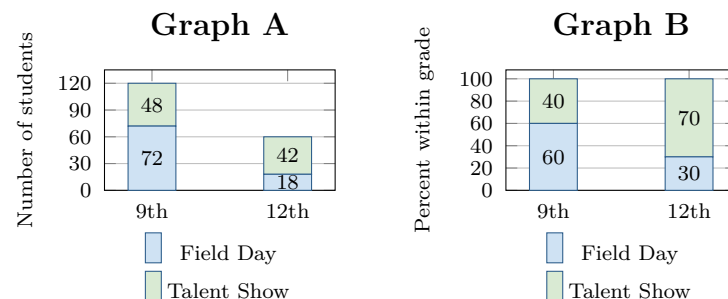
Same Responses, Different Questions

Unit 2 · Topic 2.1 · TODAY'S PROBLEM

Suppose a student council asks 120 ninth graders and 60 twelfth graders whether they prefer Field Day or a Talent Show for the spring celebration. The same responses are shown in two graphs.

Andre: *Graph B hides how many people there are because both bars reach 100%.*

Leila: *Graph A hides the relationship because the grade totals are different.*



FIRST TAKE (5 min, on your sheet)

For the question *Is celebration preference related to grade level?*, which graph would you use first? Name one visible feature.

- DOK 1** (a) From Graph A, state each grade's total surveyed and the count in each preference category.
- DOK 2** (b) Compute the conditional distribution of preference within each grade. Describe the largest contrast between the grades.
- DOK 3** (c) For deciding whether preference is related to grade level, choose the more useful graph and explain how equal-height bars settle the comparison. State what Graph A is useful for. Then give one question that Graph A cannot answer *directly from bar lengths alone* and one question that Graph B cannot answer from the display.

Video + follow-along: [u2_lesson2_live.html](https://www.khanacademy.org/a/u2_lesson2_live_html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

